

Melbourne Airbnb Market Screening

Interim Project Report

CMCE30005 Business Analytics Challenge | Semester 2 2026 | TheNextChapter Group 2

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Repository: <https://github.com/Qihang-cpu/CMCE30005-TheNextChapter>

Introduction

Our client leases residential properties and runs them as Airbnb accommodation (“rental arbitrage”), which needs lease and council permission. With limited start-up funds, the client must choose which Melbourne areas and property types to investigate before signing leases; a poor choice ties up money and delays the business.

We use the Inside Airbnb Melbourne snapshot of June 2026 supplied through the subject’s LMS (Inside Airbnb, 2026): property details, locations and dated guest reviews. Dated reviews are a countable record of past guest activity; the calendar cannot separate bookings from host-blocked dates.

The shortlist helps the client decide where to spend limited search and due-diligence time before requesting lease quotations.

Problem Definition and Objectives

Research questions: (1) Which Melbourne property segments — defined by council area (LGA), apartment or house, and number of bedrooms — show the strongest historical guest-review activity? (2) Can listing attributes improve prediction of high review activity beyond segment averages? “High” means at least 30 reviews in the year before the data were collected, the top quarter of a separate reference group of comparable listings.

Hypotheses: (H1) segments differ materially in the share of listings reaching that level; (H2) a model using listing attributes ranks listings better than the segment average alone.

Objectives: a segment comparison table, a model-versus-average test and a shortlist with uncertainty ranges.

We study entire homes with one to three bedrooms (1–3BR), what a small operator can furnish and let whole; rental units and condos form the apartment group, homes and townhouses the house group. Each segment needs at least 50 listings as minimum comparison support; quoted prices of AUD 30–1,500 are an exploratory boundary; and a first review at least 365 days before collection gives at least one year of observable review history, not proof of continuous operation.

We call the share of listings reaching 30 reviews “attainment”; 30 reviews means at least 30 reviewed stays. Attainment measures review activity, not bookings, occupancy or profit.

Data Description

The dataset contains 25,728 listings and 1,026,690 reviews, collected between 17 June and 1 July 2026. Its modelled occupancy and revenue fields are built from price, minimum stay and review counts, so ranking on them would only reproduce that construction; we rank on review counts directly.

Cleaning converts price text to numbers, standardises missing values and extracts bathroom and amenity counts; price is missing for 6,553 listings and bedrooms for 4,679. The supplied last-12-months review field spans 366 dates, so we rebuilt counts over exactly 365 days from review dates (Table 1).

Table 1. Sample funnel

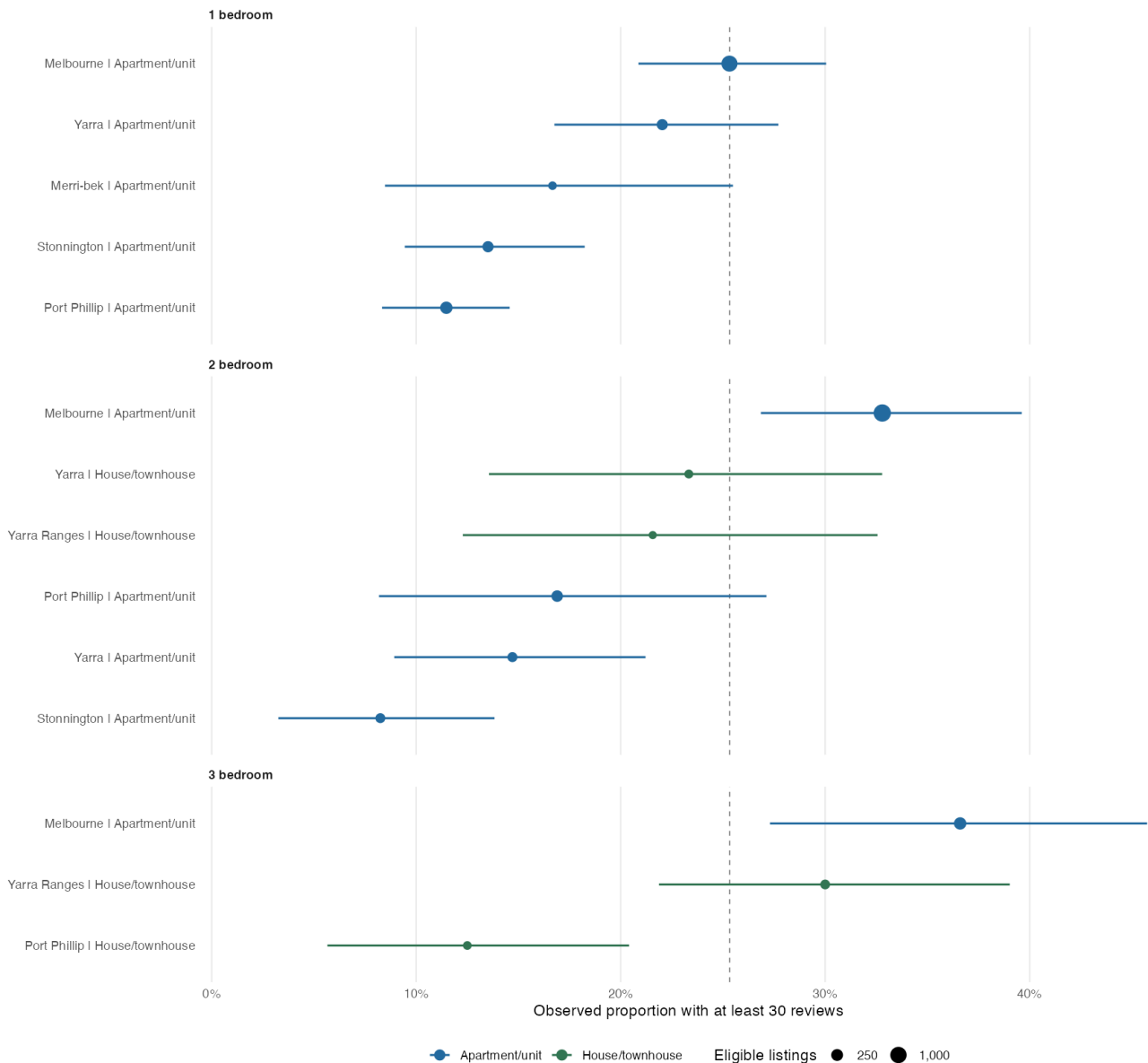
Stage	Listings
All listings in the dataset	25,728
Entire homes, 1–3 bedrooms	16,576
Standard dwelling types	14,895
Quoted price AUD 30–1,500	11,099
First review $\geq$ 365 days before	6,891

Stage	Listings
collection	
Analysis sample (50+ per segment, reference hosts excluded)	3,873
Reference group (separate, from the 6,891)	906

The reference group's 75th percentile is 29.75 reviews, giving a threshold of 30. The analysis sample (3,873 listings from 1,726 hosts) has 14 segments across six LGAs, about 64% in Melbourne LGA, and keeps 334 listings with zero annual reviews. Overall, 981 listings (25.33%) reach the threshold, and attainment ranges from 8.25% to 36.60% across segments (Figure 1).

Review activity across established residential segments

Common upper-quartile target: at least 30 reviews | 3,873 analysis listings in 14 segments



Dots: observed proportions. Lines: pointwise 90% host-cluster bootstrap intervals (1,000 draws). Dashed line: pooled rate, 25.33%. Each segment has at least 50 eligible listings; zero-review listings are retained. First review at least 365 days before each listing's scrape date. Benchmark hosts supply the cutoff and are excluded.

Figure 1. Share of listings reaching 30 reviews by segment; lines are 90% host-resampled ranges, dashed line the pooled 25.33% rate.

Of the 6,801 listings with no usable price (missing or outside AUD 30–1,500), 82% had no review in the past year: the priced sample is the more active part of the market. Listings that left Airbnb before collection are absent, which limits generalisation to new entrants.

Methodology and Analytical Approach

A fixed rule on host identifiers sets aside about 20% of hosts as the reference group that defines the 30-review target; they are excluded from all training and testing. We use five-fold validation, each part tested in turn on a model trained on the other four, with each host's properties kept in one fold so a host with many near-identical apartments cannot sit in both training and test data (scikit-learn developers, n.d.). Missing-value treatment and category encoding are learned from training data only.

Logistic regression is the simple reference; random forest and gradient boosting capture nonlinear patterns. Basic features are location, configuration, capacity, bathrooms and amenity count; an enhanced set adds coordinates, beds and nine amenities. We compare six variants of these models, with and without probability calibration and with and without current price and minimum stay. Review-based predictors, ratings and host badges are excluded.

Every model is compared with a segment-average baseline: each listing gets its segment's attainment among training hosts. AUC is how well the model orders listings that reach the target above those that do not; 0.5 is a coin toss. Average precision summarises precision across prediction thresholds. Brier score is how far its percentages are from what happened; lower is better. R handles cleaning and tables; Python (scikit-learn 1.7.2) runs the models.

Because candidates were compared after seeing fold results, the selection is repeated inside each outer training fold (nested, host-grouped), so outer folds never influence the choice; property details may also have changed during the outcome year.

Analysis Plan and Progress to Date

Following Week 5 feedback, we dropped the earlier rent-versus-revenue (ROI) design, which needed external rental data, and now compare only within the snapshot. Cleaning, review reconstruction, descriptive analysis and validation are complete.

Table 2. Predictive performance (same sample and folds)

Predictor	AUC	Average precision	Brier score
Segment average (training hosts)	0.581	0.285	0.1868
Basic property-only logistic	0.573	0.285	0.1886
Enhanced property-only random forest	0.640	0.350	0.1813

The enhanced random forest beats the segment average in all five folds and, across 20 further host splits, on AUC every time and on Brier in 16. A nested check that re-selects the model inside each outer training fold chooses the forest every time and leaves the gain unchanged: AUC +0.059 (host-resampled 95% range +0.02 to +0.10), Brier −0.0055 (−0.010 to −0.001). Its predicted rates track observed rates within two points across five probability bins. Models using current price and minimum stay reach AUC 0.749 but rely on present-day settings, so they stay out of the shortlist.

Table 3. Initial search candidates (forest predictions from models trained without each listing's host)

Segment	Listings / hosts	Observed attainment	Predicted probability	95% range
Melbourne 3BR apartments	306 / 139	36.60%	34.45%	33.0–35.8%
Melbourne 2BR apartments	1,235 / 482	32.79%	31.41%	29.6–33.0%
Yarra Ranges 3BR houses/townhouses	90 / 77	30.00%	27.82%	26.1–29.3%

The ranges resample hosts with predictions held fixed, exclude model refitting, selection and threshold uncertainty, and are much narrower than Figure 1's observed-rate ranges. Descriptive evidence supports H1: segments differ (8.25% to 36.60%), though neighbouring segments overlap. H2 is supported by nested host-grouped validation on ranking metrics but not on the shortlist: the model changes no candidate, so segment comparison carries the decision and the model adds a supplementary historical score. Descriptively, the same three segments lead under support rules of 30, 50 and 75 listings and without the history or price rule; the forest's top three recur in 19 of 20 host splits.

An out-of-time check (2,657 listings, 1,208 hosts, 10 segments, two years of reviews) uses two non-overlapping 365-day windows and a 34-review target set from the reference hosts' earlier window. The earlier year's top three segments reached it in 27.1% of listings the following year against 14.9% elsewhere, a 12.3-point lead (host-resampled 95% range 5.1–20.1); the exact top-three set recurred in 51.5% of resamples. This is a persisting historical advantage, not a new operator's profit.

Before signing, the client must still verify permissions, rent, costs and availability.

Remaining work:

1. Refit the fixed forest and re-rank with uncertainty ranges under wider property-type and history rules — Eric Huang (modelling lead) and Maksym Xu (data lead), Weeks 9–10 (by 2 Oct).
2. One-page client brief with search priorities and pre-lease checklist — Loc Le (report lead), Week 11 (by 13 Oct).
3. Final report and presentation — all, Qihang Sun coordinating, Weeks 11–12.

If model gains do not persist, segment rates remain the main screening tool; if rankings change under wider dwelling definitions, we will present a broader candidate set.

References

Inside Airbnb. (2026). Melbourne listings, calendar and reviews [June 2026 dataset supplied through CMCE30005 LMS].

Scikit-learn developers. (n.d.). Cross-validation and probability calibration.
https://scikit-learn.org/stable/modules/cross_validation.html

OpenAI. (2026). ChatGPT/Codex assistance with project planning, report drafting and code review [Generative AI outputs, September 2026].

Anthropic. (2026). Claude Code assistance with analysis-code development [Generative AI outputs, September 2026].

AI Use Acknowledgement

We used OpenAI ChatGPT/Codex and Anthropic Claude Code to support project planning, analysis-code development, interpretation of results, report drafting and English editing. This included refining the research question and reviewing the validation approach. AI-assisted text and code are included in the project. The group is responsible for the accuracy and content of the final submission.